

Pranavi Kondapalli

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EDUCATION

Carnegie Mellon University, School of Computer Science

Bachelor of Science in Artificial Intelligence, GPA: 3.4

Pittsburgh, PA

August 2023 – May 2027

EXPERIENCE

Machine Learning Researcher

May 2026 – Present

Machine Learning for Speech Processing Lab, CMU

Pittsburgh, PA

- Develop a class augmentation framework to scale speaker datasets by generating synthetic classes using adversarial mixup in embedding space.
- Design and implement an adversarial training pipeline to minimize distributional gaps between real and synthetic embeddings, improving robustness and generalization of speaker representations.

AI Software Engineer Intern – Human-Centered AI

March 2026 – Present

AI-CARING Institute

Remote

- Build a socially aware conversational AI system using value based RAG to infer latent user values from personal narratives and enable context aware responses.
- Implement a hierarchical RAG architecture with knowledge graph construction from dialogue, improving structured reasoning and long term contextual grounding.
- Benchmark LLM performance on 2,000+ narratives, validating narrative based value inference as a scalable approach for user-centered AI systems.

AI Researcher – Computer Vision and Robotics

May 2024 – March 2026

Reliable Autonomous System Lab, CMU

Pittsburgh, PA

- Built a real-time 3D computer vision pipeline using RGB-D sensors, Python, and OpenCV for multi-object detection, segmentation, and tracking in dynamic environments.
- Engineered an occlusion robust multi-object tracking and 3D localization system leveraging human pose estimation and interaction based state inference to maintain object identity across views.
- Evaluated system performance across controlled and real-world settings, achieving P=0.89, R=0.93 and demonstrating strong real-world transfer.

Software Developer

October 2023 – October 2024

ScottyLabs, CMU

Pittsburgh, PA

- Built backend systems for a campus-wide platform used by 1000+ students, implementing search and matching algorithms to streamline lost and found workflows

PROJECTS

Think Before You Trust – LLM Security Detection Framework | *Python, Llama2, AI Security*

April 2026

- Developed an adversarial evaluation pipeline analyzing prompt injection attacks and Chain-of-Thought reasoning behavior across LLM systems, leveraging 1,400+ generated prompt traces.
- Built binary classifiers on CoT embeddings using SVM, Logistic Regression, and Random Forest models, achieving up to 0.974 AUC in detecting injected vs. non-injected prompts.
- Evaluated prompt injection robustness under CoT and non-CoT conditions, demonstrating that CoT reasoning significantly reduced attack success rates across multiple prompt leaking attacks.

SteelCityBot – RAG Chatbot | *Python, FAISS, sentence-transformers, BM25, BeautifulSoup, Hugging Face*

Feb 2026

- Developed a retrieval-augmented generation system combining dense + sparse retrieval with LLM generation for domain-specific QA.
- Implemented document chunking, retrieval pipelines, and evaluation using F1 / ROUGE-style metrics.

Systems Projects | *C, Memory Management, Cache Optimization, Systems Programming*

March 2024

- Built a cache simulator in C with LRU eviction and configurable cache parameters.
- Engineered a dynamic memory allocator in C using explicit free lists and boundary tag coalescing.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript/TypeScript, SQL, R, HTML/CSS, Assembly

Frameworks/Libraries: PyTorch, TensorFlow, OpenCV, ROS, Node.js, React.js, Firebase

AI/ML: Multimodal Learning, Computer Vision, Object Tracking, Natural Language Processing, LLMs, Retrieval-Augmented Generation (RAG), Representation Learning, Model Evaluation & Prompting

AI Tools: HuggingFace Transformers, FAISS, BM25, Sentence-Transformers, Grounding DINO, SAM, RapidFuzz

Cloud/DevOps: Docker, Git, REST APIs, AWS